Standards & Guidelines Compliance

1. Electrical Safety Standards:

Battery and charging:-

Standard / requirement	Applies to	Status	Evidence
IEC 62133 – safety requirements for portable sealed secondary lithium cells and batteries	LiPo 3.7V 500mAh cell	To confirm	BOM requires the listing to state a built-in protection circuit (PCM); confirmed on arrival by reading the pack label, not yet confirmed from a listing.
Overcharge protection: cut-off at 4.2V±	TP4056 charging module	Verified	BOM: "end-of-charge 4.2V" and "1.5% charging accuracy," confirmed in the Robu.in listing text.
Overcharge / overdischarge / overcurrent protection IC	DW01A protection IC + 2x MOSFET	Verified against datasheet	Datasheet (Fortune Semiconductor): overcharge 4.3V±50mV, overdischarge 2.4V±, overcurrent ~150mV sense threshold. BOM confirms the TP4056 module listing states "DW01 protection."

Short-circuit protection	TP4056 module (protected variant)	Verified	BOM: "short-circuit/over-current protection" stated.
Charger input rating: standard USB 5V	TP4056 USB-C input	Verified	BOM: input 4.5-5.5V, matching standard USB-C 5V supply; device should only be charged from a basic 5V/1A source, not a fast-charger operating in a higher-voltage mode.

Low-voltage logic and regulation:-

Standard / requirement	Applies to	Status	Evidence
Regulated logic supply within component-rated input range	AP2112K-3.3 LDO	Verified against datasheet	Datasheet: input range 2.5-6.0V, fixed 600mA 3.3V output, built-in overcurrent and thermal-shutdown protection. Battery range (3.0-4.2V) sits inside this window at all charge states.
All sensor logic operated within manufacturer-rated voltage	MAX30205, AHT20, MAX30102, MPU6050	Verified against datasheet	All four confirmed to accept 3.3V supply per their individual datasheets.

ESP32 GPIO absolute maximum voltage (3.3V) not exceeded	Battery-sense voltage divider (2x 100kW)	Verified by design	A fully charged battery (4.2V) is halved by the two-resistor divider before reaching GPIO34, giving 2.1V, which is inside the ESP32's 3.3V absolute maximum input rating.

2. Component-Level Compliance:

Component	I2C address	Required external parts	Status
MAX30205 (skin temperature)	0x48 default (all address pins low)	4.7kW SDA/SCL pull-ups (shared bus); 0.1µF decoupling cap on VDD	Design verified; module-level pull-ups/cap not stated by seller, carrier board supplies its own pull-ups and cap regardless.
AHT20 (ambient temperature)	0x38 fixed	10kW pull-ups typically onboard	Address verified against datasheet; onboard pull-ups not confirmed from a specific listing, to check on arrival.
MAX30102 (heart rate / SpO2, pulse only used)	0x57 fixed	Onboard 1.8V regulator; pull-ups on SDA/SCL	Address verified. Known risk: some modules wire pull-ups to the internal 1.8V rail instead of 3.3V, which stops the chip being detected on a 3.3V bus. To be confirmed by I2C scan on arrival.

MPU6050 (motion / wear detection)	0x68 default (AD0 low)	Onboard voltage regulator, pull-ups, AD0 tied low	Verified in listing (onboard LDO, AD0 low by default on GY-521 modules); carrier board additionally ties AD0 to GND regardless of module default.
ESP32 DevKit V1 (controller)	n/a	Onboard CP2102 UART bridge, USB-C port, EN pull-up, decoupling caps: all pre-fitted by manufacturer	Verified in listing (CP2102, Type-C present). Module marking to be checked on arrival to confirm it is the classic ESP32-WROOM-32 silicon, not a newer S2/S3/C3 variant with a different pinout.

3. Mechanical and Enclosure Guidelines:

Guideline	Applies to	Status	Evidence
Rounded corners / no sharp edges on a body-worn enclosure	3D-printed PETG pod	Verified by design	Enclosure specified with 3mm rounded corners; no exposed sharp edges in the CAD model.
Skin-contact material should be non-irritating for prolonged overnight wear	TPU/silicone skin liner; copper pad + foam at the sensor window	Verified by design	TPU and silicone are standard skin-safe materials used in consumer wearables; any surface touching skin is lined per the mechanical structure spec.

Battery physically isolated from the skin-contact side where possible	Battery pocket placement	Verified by design	Battery pocket sits under the main board, on the outward side of the enclosure, away from the skin-contact sensor window.
Adjustable strap to avoid excessive tightness (circulation risk) across a range of arm sizes	Elastic nylon strap, 35mm, hook-and-loop	Verified by design	Strap specified as adjustable across 22–34cm inner upper-arm circumference.

4. Design Decisions With a Compliance Basis:

• Voltage divider ahead of GPIO34: added specifically because the ESP32's GPIO absolute maximum is 3.3V, while a freshly charged LiPo reaches 4.2V, as connecting the battery directly would exceed the pin's rated input and risk permanent damage.

• AP2112K chosen with input range 2.5-6.0V: selected so the full battery discharge range (4.2V down to the DW01A's 2.4V cut-off) stays within the regulator's safe input window at all times.

• Sleep-only sensing window (overnight wear, not continuous): reduces continuous skin contact time and battery duty cycle versus an always-on wearable, lowering both thermal and electrical exposure during use.